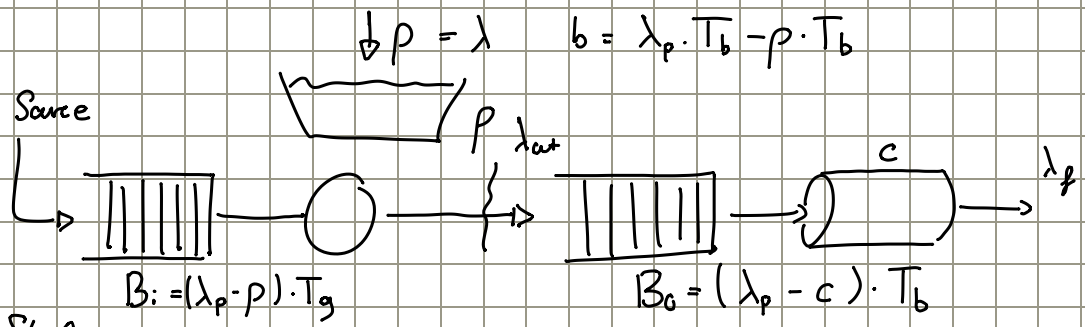
23/10/2026

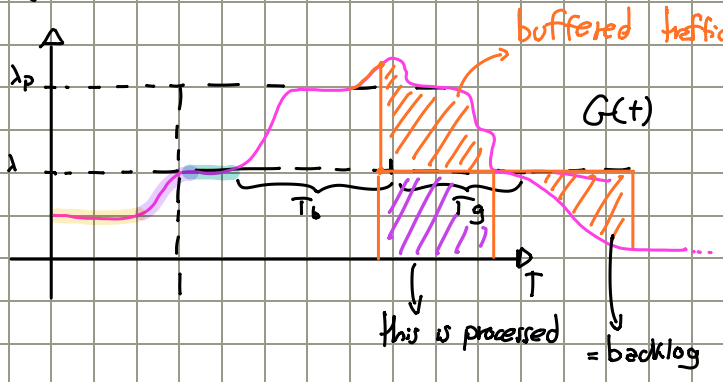
Little recap from lesson 7



SLA

- λ_a: long term average rate
- $\lambda_p > \lambda_a$: peak rate
- T_b: burst duration to be forwarded through the TBF at peak rate
- Grace period T_g: max time during which a source can generate traffic exceeding the shaping without causing packet loss

Question:

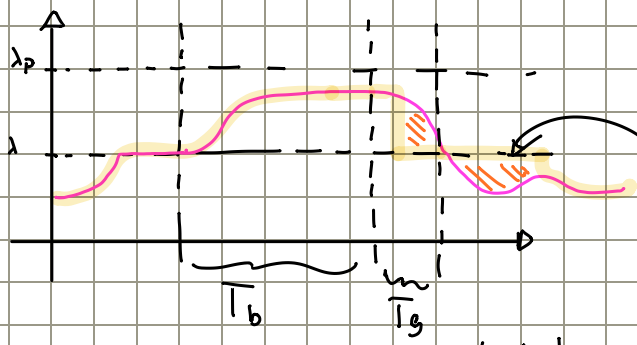


G(t)

What we will expect? Come cambia l'andamento?

We are transmitting tokens more than λ_p, so we are consuming tokens faster, so at a certain point before T_g, we will have a drop

Other scenario, same setting:



Always under λ_p

until areas are equal

$$\frac{b}{\tilde{\lambda} - \rho} \rightarrow T_b \triangleq \frac{b}{\lambda_p - \rho}$$

Burst period can be exceeded if...

We consider some practical case

Source

Sensor node that repats a data packet of 512 bytes every minute

Randomly the node enters an alarm state that lasts for $D = 30''$ it generates CBR traffic

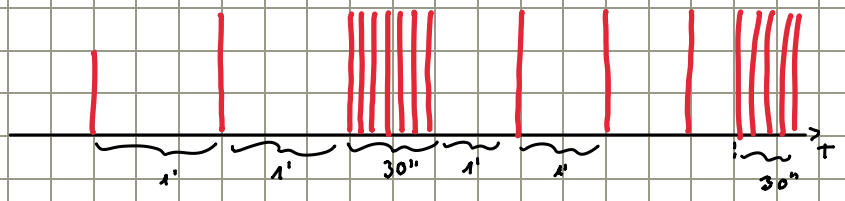
$$A_i = 1'', \quad B_i = 1, \quad W_i = 512\ B$$

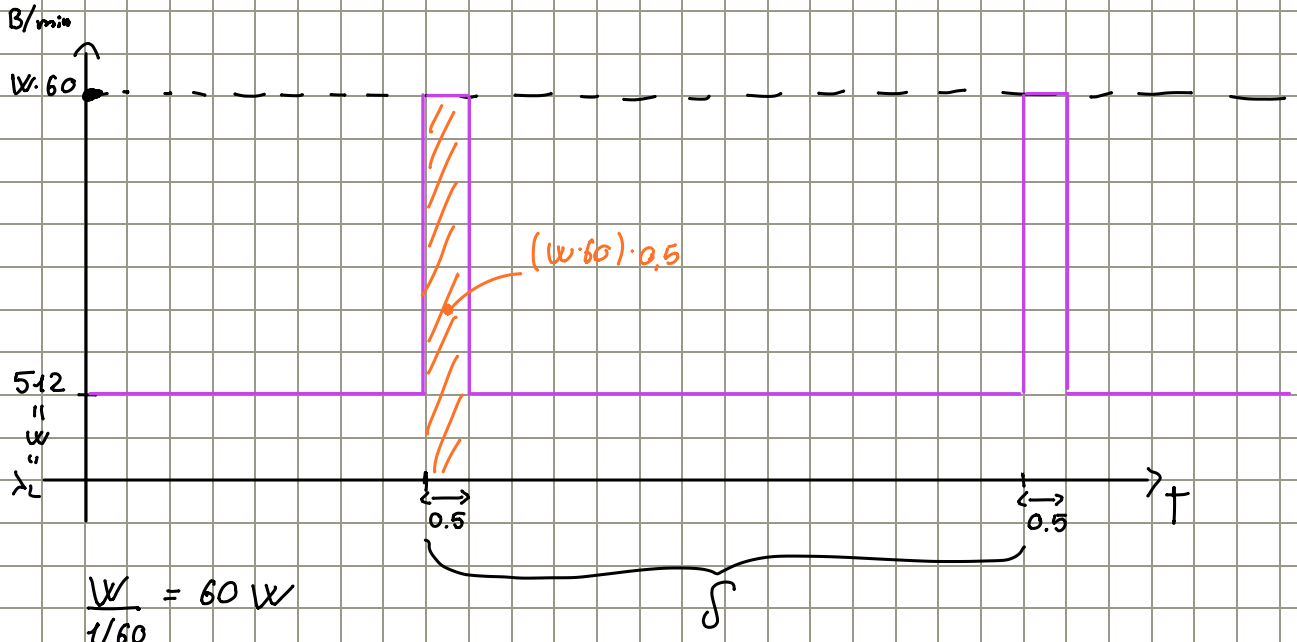
Alarms occur as P.P.P. of rate μ [alarm/minute]

SLA = ?

$\lambda = ?$ $\quad T_b = ?$ $\quad \Rightarrow \quad \rho, b, B_i \rightarrow TBF$

$\lambda_p = ?$ $\quad T_g = ?$ $\quad\quad C, B_o \rightarrow link$





$$\frac{W}{1/60} = 60\,W$$

$$\lambda = \frac{E[\#\text{ profiles in } \delta]}{E[\delta]} = \frac{E[W \cdot 60 \cdot 0.5 + (\delta - 0.5) \cdot W]}{E[\delta]}$$

$$= \frac{512 \cdot 30 + 512 \cdot (E[\delta] - 0.5)}{E[\delta]} =$$

$E[\delta] = \frac{1}{\mu} = 10$

$$\stackrel{!}{=} \frac{512 \cdot 30 + 512 \cdot 9.5}{10} \approx 2048 \text{ Bytes/min}$$

$$= 2022{,}4 \text{ Bytes/min}$$

$\lambda_p = w \cdot 60 = 512 \cdot 60 = 30720$ Bytes/min

$T_b = 0{,}5$ min

$b = T_b(\lambda_p - \rho) = 0{,}5(30720 - 2022)$ tokens
($\rho \approx \lambda$)

Do we need a grace period? Yes, because the arrivals are random so ... (finish explanation)

At time $t=0$ the sensor exits the alarm state $b_i(t) = 0$

(Input buffer is empty) $b_i(t) = 0$

(Token bucket is empty) $b(t) = 0$

? Prob. that next alarm will not be at least partially buffered?

$$= \mathbb{P}[b(\delta) < \underbrace{\underbrace{(w \cdot 60 - \rho)}_{\text{Peak rate}} \cdot \underbrace{0{,}5}_{T_b}}_{b}] = \mathbb{P}[\underbrace{(\lambda}_{\rho} - \underbrace{\lambda_L)}_{w/1'} \cdot \delta < b]$$

$$= \mathbb{P}\left[\delta < \frac{b}{\lambda - \lambda_L}\right] = e^{-\mu \frac{b}{\lambda - \lambda_L}} = e^{-\mu \frac{\lambda_p - \lambda}{\lambda - \lambda_L} \cdot T_b}$$

⇒ could we change something?

No.

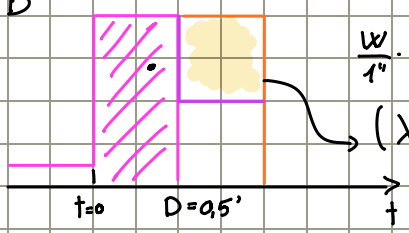
$b = (\lambda_p - \lambda) T_b$

How many bytes will be buffered?

We can say it on average (mean number of bytes?)

What's the PDF of # of

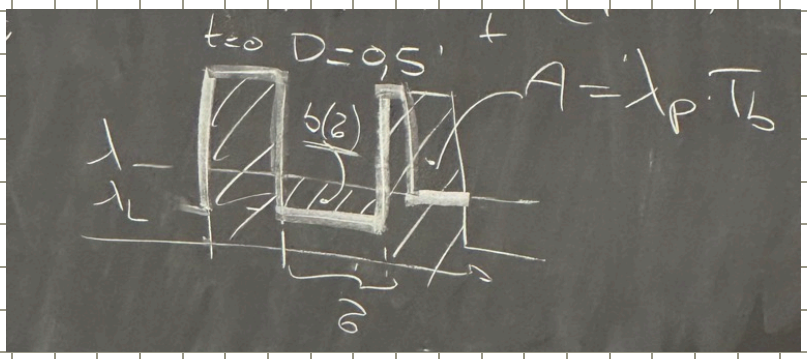
$b(\delta) \leq b$



$\frac{w}{1'} \cdot D = (60w) \cdot D = \lambda_p \cdot T_b \triangleq A$

$(\lambda_p - \lambda) \cdot T_b < A$

Prof uses PDF also for DRV



$$\frac{b(\delta)}{\lambda - \delta} = T' = \text{time to consume my credits}$$

If $T' \geq T_b \Rightarrow$ no buffer

$T' < T_b \Rightarrow$ buffering

$\beta(\delta)$ = # of bytes buffered if next alarm is at time δ

$$= \max\left[\left(\lambda_p - \lambda\right)\left(T_b - \frac{b(\delta)}{\lambda_p - s}\right); 0\right]$$

$$P[\beta(\delta) > a] = P\left[\left(\lambda_p - \lambda\right)\left(T_b - \frac{b(\delta)}{\lambda_p - p}\right) > q\right] = P[b(\delta) < f(a)]$$

$$= P\left\{\delta < \frac{f(a)}{p - \lambda_L}\right\}$$